

DAVISON AAF, VA, USA

WMO: 724037

Lat: 38.717N

Lon: 77.183W

Elev: 22

StdP: 101.06

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.5	-19.3	0.7	-7.2	-16.5	0.9	-4.3	11.9	0.3	11.0	0.1	2.6	310	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.2	24.0	33.7	23.5	32.3	23.1	25.9	31.7	25.1	30.8	24.5	29.7	3.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.3	28.3	23.6	18.5	27.6	22.9	17.7	26.8	79.9	32.2	76.4	30.8	73.8	29.8	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.2	6.9	-14.7	37.0	3.5	1.9	-17.3	38.4	-19.3	39.5	-21.3	40.6	-23.8	42.0		
(5)			-15.0	27.5	3.2	1.2	-17.3	28.4	-19.1	29.1	-20.9	29.8	-23.2	30.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	1.7	3.8	7.7	13.8	18.5	23.3	25.7	24.6	21.1	14.7	8.2	4.4	(6)
(7)		DBStd	9.38	5.61	5.50	5.29	4.57	4.23	3.14	2.62	2.71	3.75	4.36	4.49	4.95	(7)
(8)		HDD10.0	851	264	183	108	14	1	0	0	0	8	85	187		(8)
(9)		HDD18.3	2337	515	406	331	149	52	3	0	16	129	303	433		(9)
(10)		CDD10.0	2320	8	11	38	129	263	398	488	453	332	154	32	13	(10)
(11)		CDD18.3	765	0	0	2	14	57	152	229	195	97	17	0	1	(11)
(12)		CDH23.3	7594	0	3	38	196	601	1494	2404	1873	836	141	6	3	(12)
(13)	CDH26.7	3163	0	0	10	57	212	605	1136	809	301	33	0	0	(13)	
(14)	Wind	WSAvg	2.2	2.7	2.8	3.1	2.6	2.0	1.9	1.7	1.6	1.7	1.8	2.2	2.4	(14)
(15)	Precipitation	PrecAvg	1057	75	63	91	80	94	105	101	92	103	84	62	85	(15)
(16)		PrecMax	1283	158	172	218	131	216	279	185	186	316	186	128	165	(16)
(17)		PrecMin	809	32	15	0	43	24	34	41	36	0	0	7	40	(17)
(18)		PrecStd	124	30	34	45	29	47	57	38	36	69	45	38	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	22.8	27.5	31.2	33.9	36.0	37.8	37.0	35.1	30.5	23.9	21.2	(19)
(20)			MCWB	16.2	15.5	16.3	19.0	22.2	24.3	24.4	23.6	22.0	22.2	16.4	15.8	(20)
(21)		2%	DB	17.3	18.7	23.2	28.1	31.4	33.7	35.7	34.2	32.3	27.1	21.0	17.9	(21)
(22)			MCWB	13.4	12.4	14.3	17.6	21.6	23.7	24.3	23.4	22.4	20.4	15.3	14.7	(22)
(23)		5%	DB	13.8	15.6	20.2	25.3	29.0	32.1	33.9	32.6	30.2	24.6	18.4	15.3	(23)
(24)			MCWB	9.9	10.1	12.8	16.4	20.9	23.2	24.2	23.2	22.0	18.8	13.4	11.9	(24)
(25)	10%	DB	10.2	12.5	17.2	22.8	26.9	30.3	32.3	31.1	27.9	22.5	16.2	12.3	(25)	
(26)		MCWB	6.8	8.0	11.1	14.9	19.3	22.4	23.7	22.7	20.7	17.5	11.8	8.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	16.6	17.8	20.8	24.3	26.3	27.6	26.5	25.2	24.5	18.8	17.7	(27)
(28)			MCDWB	19.2	21.7	24.5	26.1	30.4	32.6	33.7	32.3	29.9	27.1	21.4	19.5	(28)
(29)		2%	WB	14.2	13.9	16.0	19.5	23.1	25.1	26.2	25.4	24.1	22.0	16.7	15.4	(29)
(30)			MCDWB	17.3	17.7	21.5	25.5	29.0	31.2	32.3	31.0	28.6	25.4	19.6	17.9	(30)
(31)		5%	WB	10.3	10.8	14.0	17.9	21.9	24.3	25.4	24.7	23.2	20.5	14.7	12.6	(31)
(32)			MCDWB	12.9	13.6	18.8	23.8	26.9	30.0	31.3	29.9	27.3	23.5	17.4	15.0	(32)
(33)	10%	WB	7.1	8.6	11.6	16.4	20.8	23.6	24.7	23.9	22.3	18.5	12.8	9.5	(33)	
(34)		MCDWB	9.8	12.4	16.2	21.6	25.0	28.7	30.0	28.7	26.4	21.3	15.5	11.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	10.6	11.6	13.1	12.3	11.8	11.5	11.4	11.5	12.2	11.6	9.5	(35)
(36)			MCDBR	14.1	15.1	16.7	16.8	15.5	13.7	13.6	13.5	13.7	14.3	14.8	12.9	(36)
(37)			MCWBR	10.6	10.1	9.5	8.0	7.2	5.7	4.9	4.7	5.6	8.0	9.3	9.9	(37)
(38)		5% WB	MCDBR	12.5	13.2	14.6	14.2	12.9	11.8	11.3	11.1	10.9	11.7	11.4	11.7	(38)
(39)			MCWBR	10.7	10.0	9.6	8.0	6.5	5.6	5.1	4.6	5.1	7.2	8.7	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.322	0.356	0.404	0.443	0.501	0.522	0.517	0.428	0.360	0.324	0.309	(40)	
(41)		taud	2.494	2.433	2.399	2.284	2.204	2.082	2.043	2.039	2.263	2.457	2.518	2.522	(41)	
(42)		Ebn at Noon	874	904	903	874	843	791	770	762	815	848	846	844	(42)	
(43)		Edh at Noon	78	95	109	130	143	162	167	164	124	92	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.11	2.94	3.92	4.92	5.57	6.13	6.00	5.36	4.37	3.30	2.40	1.86	(44)	
(45)		RadStd	0.18	0.33	0.28	0.37	0.61	0.46	0.31	0.28	0.51	0.37	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	+78

Nomenclature: See separate page